

Fine-Tuning a Language Model on Its Own Verified Lean Proofs: Transfer, Generalisation and Saturation Across Three Iterations

Roberto I. Ono Filho
Independent researcher
`ono.roberto@gmail.com` ORCID [0009-0006-8650-629X](https://orcid.org/0009-0006-8650-629X)

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Abstract

We ask whether a language model improves on problems it has not seen when the proofs it produced and a verifier accepted are fine-tuned into its weights, with no human selecting or editing them. The setting is a ladder of six graded families of small Lean 4 worlds whose instances differ in constants, not in structure. The loop ran for three iterations on a served 27B model; every stage was read once against a pre-registered rule, and every adapter was compared with one trained on the generator’s template proofs of the same cells. Template proofs write their form into the weights: their adapter solves its trained steps at the first submission and no others, and writes skeletons with guessed coefficients that reasoning in context fills. The model’s own accepted proofs transfer within their steps and, at the second iteration, solve two steps for which no template was shown; the loop stops at L4, whose invariant the lineage produced twice in 48 sessions and the second adapter never writes there. On four new families the loop’s adapter rises over the base by the rule in three, at a seventh to two fifths of the tokens, while the template adapter of the same cells writes its nearest trained skeleton and does not fall below the base. A third iteration raises how often the needed kind of invariant is tried first and leaves the solve rate where it was; a registered composition test did not fire. A hand-validated classifier of the kind tried first (3,660 sessions) separates prior from execution: consolidation moved the prior toward forms in the training data; thinking and practice improved execution and repair; in three of the four new families the loop’s adapter guessed the needed kind less often than the base and solved more by repairing after a rejection. Every family an adapter solved, the base also solved at eight attempts and 131k thinking tokens; at the level of instances and small budgets the adapters cover more; support at larger budgets was not measured.

1 The question

The previous papers of this series measured what an office-scale language model does with a stimulus and with an idea it is handed [23, 24, 26, 27], and concluded that the model applies an idea it is handed in operable form but did not, in those programs, produce an operator of its own [25]. Weight consolidation appeared there once, in the bin-packing lineages of the second paper, where it moved the mass of proposals toward the known good and thinned the tails [24]. This paper asks the question the series left open: when nothing is handed to the model and its only training data are its own verifier-accepted proofs, does it improve on instances it has not seen, and where does the improvement stop?

“Without selection” means here that no human selects, edits, ranks or orders the proofs that enter training; every accepted proof of a production run enters as it came. The humans still supply the generator of problems, its six graded steps, the verifier, the choice of which steps each

iteration attempts, and the decision to retrain from scratch on the cumulative set. Automatic selection of training rationales is also how STaR, ReST-EM and rejection-sampling fine-tuning work [35, 44, 47], so the phrase describes the data regime only. The design’s distinctive element is its matched control. A generator produces sibling instances of a formal problem, a verifier decides, and for every instance a template reference proof exists, so that each self-trained adapter can be compared with an adapter trained on the reference proofs of exactly the same cells. The program was run as eight pre-registered studies (“nodes”) between 13 and 20 September 2026, each archived with a SHA-256 manifest (Table 1; Figure 2 shows which arm was trained and read where). Each of the seven confirmatory nodes was read once by a script written before its data (A8 also had a declared second read, Section 5); the descriptive node A6 registered its method and questions before it ran, and its scripts were committed with its first read and rerun twice as later nodes added sessions. The program’s findings, stated without numbers here, with them in the sections that follow, and summarised in Figure 1:

1. **Template proofs write their form into the weights** (Section 3). An adapter trained on the reference proofs of two steps solves those steps at the first submission and no others. The six-form adapter solves the four steps whose measure is read off the moves and fails the two whose parameters must be inferred, writing the right skeleton with guessed coefficients. Reasoning in context fills the coefficients; the same change of sampling configuration without reasoning leaves the rate where it was.
2. **The loop closes without selection and climbs** (Section 4). The base’s own accepted proofs, consolidated as they came, transfer within their steps. The resulting adapter, thinking, produces proofs of two higher steps for which no template was ever shown, and their consolidation solves those steps at the first submission without thinking. The loop stops at L4, the one step whose invariant, a product of two different fields in a three-field world, the lineage produced twice in 48 sessions and the adapter never writes there.
3. **On four new families the loop’s adapter rises over the base and the template adapter writes its nearest trained skeleton** (Section 5). The loop’s adapter rises over the base in three of the four families by the rule and in all four descriptively, at a fraction of the tokens. The template adapter trained on the same cells does not fall below the base, but writes in every first submission the trained skeleton nearest to the world, and combines two skeletons far less often than the loop.
4. **A third iteration saturates** (Section 6). Adding the adapter’s own proofs of the new families raises how often the needed kind of invariant is tried first and leaves the solve rate where it was; it does not make the new families solvable without thinking; the registered composition test did not fire.
5. **What is tried and what is executed can be separated** (Section 7). A classifier of the kind of invariant in the first submission, validated by hand, shows that consolidation moved the prior over forms and that thinking moved execution and repair. In three of the four new families the loop’s adapter guesses the needed kind at the first submission less often than the base, and it solves more because it repairs its guess after the verifier rejects it.

2 Objects, arms and rules

Terms. An *iteration* is one turn of the loop (produce, consolidate, read). A *session* is one attempt at one instance by one arm; inside it the model makes up to twelve *submissions*. An

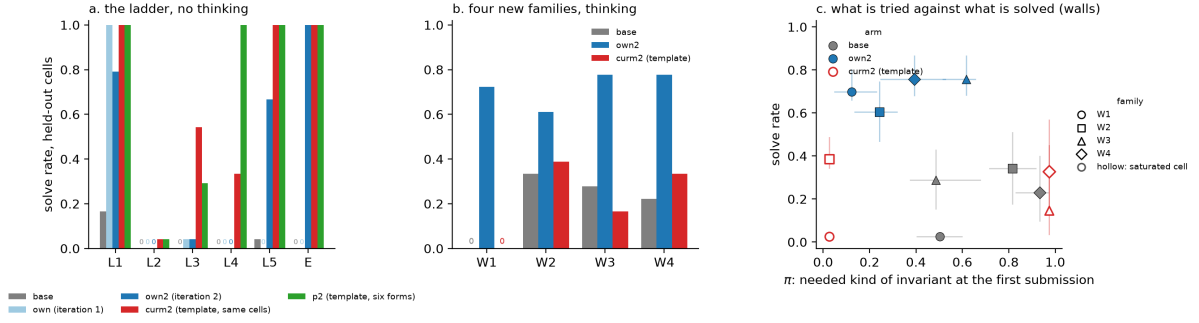


Figure 1: The three results in one figure, from the read files; a zero-height bar is printed as 0. (a) Held-out solve rates on the ladder without thinking: the base, the two self-trained adapters (own, own2), the template adapter trained on own2’s cells (curr2) and the six-form template adapter (p2). (b) The four new families with thinking: base, own2 and the template adapter of the same cells. (c) On the walls, what is tried against what is solved: π , the rate of the needed kind of invariant at the first submission (median and 90% interval), against the solve rate, per arm and family; the lines are 90% intervals, and a hollow marker is a saturated cell, where every session of every instance has or lacks the kind and π sits at the smoothing floor or ceiling (Section 7). The base and the template adapter tend to start with the needed kind and then fail to complete the proof; the loop’s adapter tends to start with another kind and reach the needed one after a rejection.

node (read)	question and registered primary	sessions, GPU	registered verdict (plain English)
A2 (14 Sept.)	does a template form write into the weights? primary: the form-specific adapter rises on its two steps and on no other	576, US\$32	form_written_and_form_specific: yes, and only that form
A2 replication, P4, E8 (14 Sept.)	new instances and adapters; does thinking complete the form? primary: same rises; the adapter with thinking rises over itself without thinking in the same configuration	768, US\$75	replicated; reasoning_is_the_active_ingredient_given_the_form
Loop, iteration 1 (15 Sept.)	do the model’s own accepted proofs move it? primary: own rises over the base on its steps, without and with thinking	528, US\$69	self_consolidation_moves: yes
Loop, iteration 2 (16 Sept.)	does the ladder climb without selection? primary: own2 rises over the base at the two produced steps	744, US\$112	ladder_climbs_without_curator: yes, without thinking
A7 (17 Sept.)	is there a step the base never passes? wall: base 0 of 48 sessions at 131k tokens	234, US\$91	not_a_wall_at_scale: no candidate was a wall
A8, A8-c (19 Sept.)	does the template adapter narrow, does the loop’s generalise? two-of-four-families rules	216 (45 of them in the second read), US\$107	self_generalises_only
A9 (20 Sept.)	does a third iteration compose (primary), cheapen, or saturate?	594, US\$58	saturates
A6 (19 Sept.)	what is tried versus what is executed	3,660 (re-read)	secondary, descriptive

Table 1: The nodes. Sessions are distinct observed prover sessions; GPU cost is the rented H200 time of the node at list price, including the training of its adapters and its failures. The pre-registration commit of each node is listed in Appendix A.

attempt count (“three attempts”) is the number of sessions per instance and arm. A *round* is one of the two rounds of the template consolidation node. An *arm* is a condition (weights and configuration); a *cell* is an instance, a step and an attempt. A *node* is one pre-registered study, and its *read* is the single execution of its registered analysis script on its complete data.

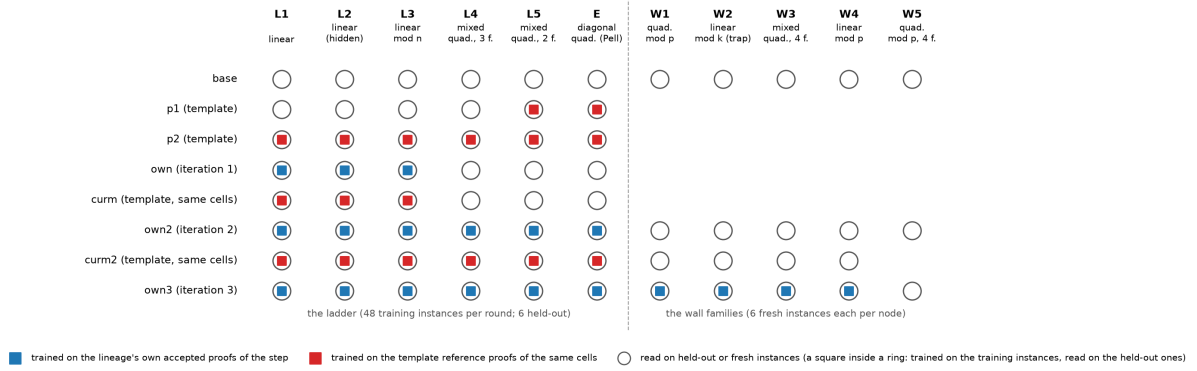


Figure 2: The design. Columns: the six ladder steps and the five wall families, each with the kind of invariant its intended proof needs (f.: fields of the state). Rows: the arms. A square marks what an adapter was trained on (blue: the lineage’s own accepted proofs; red: the template reference proofs of the same cells); a ring marks where an arm was read, on held-out instances of the ladder or fresh instances of a family.

The ladder. A world is a small Lean 4 development [6] over Mathlib [36]: a state of two to four integer fields, four moves, a reachability relation, a start state, and a target theorem stating that a family of states is unreachable. The intended proof exhibits a quantity conserved by every move (a *measure*), proves conservation by induction on reachability, and evaluates it on start and target. Six steps are graded by the conserved quantity: L1, a difference of two fields; L2, a hidden linear form $\alpha x + \beta y + z$; L3, a linear form modulo an index n ; L4, a quadratic form with offsets $(x - c_x)(z - c_z) - (y - c_y)^2$ under shears, in three fields; L5, the form $x^2 - kxy + y^2$ under the Vieta map, in two; E, a Pell form $x^2 - Dy^2$. Instances are seeded: they differ in their constants and share their structure, so “held-out” means held-out constants rather than a shift of distribution. Every instance carries a reference proof accepted by the same checker the sessions use and a mutation check (the reference is rejected on a falsified world, and a witness proves the falsified statement wrong). Six held-out instances drawn on 14 September for the replication, and never trained on, serve every ladder reading after it (the first round had its own six); 48 training instances per round serve production and training.

The template proofs. The reference proof of an instance is emitted by the generator from one skeleton per step: a **measure** definition, a step lemma, a bridge lemma by induction on reachability, and a closing tactic (**omega** for the linear and modular steps, **Int.sq_nonneg** and **grind** for the quadratic ones). Across the 288 reference proofs of a round there are six skeletons, one per step, with the constants of each instance filled in. We call the adapters trained on them *template adapters*; the arm codes keep the historical name *curm* (“curated, matched”), and where the text says “curated” it means these uniform reference proofs and nothing more general.

The walls. Four further families were built for the wall search (Section 5) in the same format: W1, a Pell form conserved only modulo a prime p under translations by p (a composition of E with L3); W2, a plane lattice whose exact linear invariant does not separate start from target, so that a form modulo the lattice index is needed (an L2 trap that requires L3); W3, a hyperbolic

form in four fields with offsets, preserved by SL_2 shears on two vectors; W4, the L3 construction with a prime between 11 and 19. A fifth, W5, was built for the third iteration: the form of W3 conserved only modulo a prime, as in W1. An exhaustive certificate checks, for every instance, that no exact linear invariant separates start from target and no linear form modulo $m \leq 30$ ($m \leq 12$ in four fields) does either, apart from the intended family. Section 6 reports two shortcut families this certificate did not cover, one of which also occurs in W1; the ladder had no such certificate (Section 10).

The session. A prover session shows the model the public files of a world and asks for a complete Lean file proving the target (Appendix C gives the prompt). Each submission is compiled and checked (namespace, statement, standard axioms only); a rejection returns the compiler’s errors; the session ends at the first accepted submission, at twelve submissions, at the output budget, or at the time limit. Two configurations are used throughout: *no thinking* (16,384 output tokens, T 0.7, top- p 0.8, 900 s) and *thinking* (131,072 tokens, T 1.0, top- p 0.95, 3,600 s, the model’s reasoning enabled). Between submissions the conversation carries only the submitted file and the errors; the reasoning of a rejected submission is not carried forward. With thinking on, the model often writes the Lean file inside its reasoning and leaves the answer empty; the file is then extracted from the reasoning (an amendment recorded before the affected cells), so a “first submission” under thinking is what the verifier received after the model reasoned. An infrastructure failure (a dropped connection) is not an observation, and the cell is rerun, at most three times.

Model and adapters. The base is Qwen3.8-27B, a 2026 release of the Qwen3 line, served from the repository `Qwen/Qwen3.8-27B` of the Hugging Face Hub [31], downloaded afresh on each rented instance between 13 and 20 September 2026 (the repository revision was not recorded). It is the post-trained release rather than a pre-trained checkpoint: the two configurations above are its thinking and no-thinking modes, and the family’s technical report [30] describes the design in which the no-thinking mode is a distillate of the thinking mode. It is served in BF16 by vLLM 0.29.0 (transformers 5.17.0) on one rented H200 (list price US\$4.59 per hour). Every adapter is a LoRA [13] of rank 16 ($\alpha = 32$, dropout 0.05) on the attention and MLP projections, trained with peft 0.20.0 on torch 2.13 with AdamW at 2×10^{-4} , cosine schedule with 5% warm-up, effective batch 8, 4,096 tokens, bf16, for exactly 108 optimiser steps, with the loss on the assistant tokens of a [user: prover prompt, assistant: proof] pair rendered through the model’s chat template with thinking disabled. The last state is saved; there is no selection among checkpoints; one training seed per adapter (2026091301 and 2026091402 for the two rounds of A2; 2026091501, 2026091502 and 2026091901 for the three iterations of the loop). Each iteration retrains from scratch on the cumulative set. The same recipe and step count serve every adapter, so that arms differ only in their data. Arm codes: *base*; *p1* (96 reference proofs of E and L5), *p2* (288 reference proofs of all six steps), *p3* (288 files with the same worlds, header and tactics on a non-inductive target, a syntax control); *own* (the 60 proofs the base produced and the verifier accepted at L1–L3, iteration 1), *curm* (the reference proofs of exactly those 60 cells); *own2* (the 135 proofs of iterations 1 and 2, cumulative), *curm2* (the reference proofs of the same 135 cells); *own3* (*own2*’s data plus the 52 proofs *own2* produced on the wall families, iteration 3). A trailing *t* or the word *thinking* marks the thinking configuration.

The rule. The unit of inference is the instance. For two arms on the same cells, the paired difference of solve rates over the six instances is tested by a one-sided sign-flip permutation ($p < 0.05$; the minimum with six instances is 0.0156), once in each direction, and a rise or fall is declared only if the pooled difference is also at least 0.25 in absolute value. Where a comparator sits above 0.75 (ceiling) or below 0.25 (floor) the rule cannot fire, and this is reported rather

than worked around. The per-contrast level is not corrected for multiplicity. The protection is that each node registered one primary contrast or a conjunction of contrasts (“rises in at least two of four families”) and a verdict vocabulary before any session ran, and was read once by a script written before its data; every other contrast in the tables is secondary and is marked as such in the captions. Descriptive quantities (token counts, first-submission solves, the forms of the submissions, the instance coverage) are reported as such. Amendments were recorded with the clock time and before the cells they affected.

3 Template proofs write the form

Table 2 gives the two rounds. In both, the form-specific adapter p1 solves E and L5 in 24/24 held-out cells, all at the first submission, all with the trained skeleton, and moves nothing else. It even loses the base’s 4/24 at L1 by writing the Pell form $x \cdot x - y \cdot y$ as a measure there (24/24 first submissions), a negative transfer between forms. The six-form adapter p2 solves L1, L4, L5 and E in 24/24 and fails L2 (0 and 1 of 24) and L3 (7/24 in both rounds). In those two steps it writes the right skeleton with guessed coefficients ($1 \cdot x - 2 \cdot y + z$ where the world’s form is, say, $-x - y + z$; c and n guessed in $(x + cy) \bmod n$): it has learned the form of the proof but not how to derive the coefficients from the moves. The syntax control p3 solves nothing in 288 sessions. The replication, with new training instances, new held-out instances and adapters trained from scratch, reproduces every cell to within one solve (base 5 and 5 of 144; p1 48 and 48; p2 103 and 104; p3 0 and 0). Training losses fell to 3.4×10^{-3} or below, so the adapters memorised their examples, and the transfer is to new constants rather than to the examples. Section 7 qualifies “only the form”: execution moved too where the form has no parameter to infer, and at L3 it rose from 0.02 to 0.29, short of the ceiling.

arm	L1	L2	L3	L4	L5	E
<i>round 1 (13–14 September)</i>						
base, no thinking	4/24	1/24	0/24	0/24	0/24	0/24
p1: form of E/L5 (96 examples)	0/24	0/24	0/24	0/24	24/24↑	24/24↑
p2: six forms (288)	24/24↑	0/24	7/24	24/24↑	24/24↑	24/24↑
p3: syntax control (288)	0/24	0/24	0/24	0/24	0/24	0/24
<i>replication, new instances and adapters (14 September)</i>						
base, no thinking	4/24	0/24	0/24	0/24	1/24	0/24
p1: form of E/L5 (96 examples)	0/24	0/24	0/24	0/24	24/24↑	24/24↑
p2: six forms (288)	24/24↑	1/24	7/24	24/24↑	24/24↑	24/24↑
p3: syntax control (288)	0/24	0/24	0/24	0/24	0/24	0/24

Table 2: Template consolidation, two rounds, six held-out instances $\times$ four attempts per step, no thinking. ↑: rise against the base by the rule. Registered primaries: p1 on E and L5; p2 on every step; p3 on none.

Thinking fills the coefficients. Table 3 isolates what reasoning adds once the form is in the weights. With thinking on and nothing else changed (same cells, same input hashes, the only differing field of the agent being `enable_thinking`), p2 goes from 2/24 to 19/24 at L2 and from 5/24 to 18/24 at L3 (+0.71 and +0.54, $p = 0.016$); the configuration alone (sampling temperature, budget, time) moves nothing (+0.04, −0.08). In the sessions, the adapter writes the skeleton with guessed coefficients, is rejected, and corrects the coefficients in its second response with about 3,000 tokens; without thinking it repeats the guess up to the twelve-submission limit against the same error. The base with the same thinking reaches 6/24 and 8/24 at about 100,000 tokens per session, never at the first submission, with heterogeneous forms. So with the form in the weights, the remaining work, deriving the coefficients, is done

in context, and only when the model can reason over the verifier’s rejection.

arm	weights	thinking	T / top- p	budget	L2	L3
base	base	off	0.7 / 0.8	16k	0/24	0/24
basen	base	off	1.0 / 0.95	131k	2/24	0/24
baset	base	on	1.0 / 0.95	131k	6/24	8/24
p2	p2	off	0.7 / 0.8	16k	1/24	7/24
p2n	p2	off	1.0 / 0.95	131k	2/24	5/24
p2t	p2	on	1.0 / 0.95	131k	19/24	18/24
<i>paired contrasts (rate difference, sign-flip p)</i>						
p2t – p2n (thinking, same configuration)				+0.71 ($p = 0.016$)	+0.54 ($p = 0.016$)	
p2n – p2 (configuration alone)				+0.04 ($p = 0.500$)	-0.08 ($p = 0.938$)	
baset – basen (thinking, no form)				+0.17 ($p = 0.125$)	+0.33 ($p = 0.016$)	
p2t – baset (form, both thinking)				+0.54 ($p = 0.016$)	+0.42 ($p = 0.016$)	

Table 3: The form and the reasoning, L2 and L3, six held-out instances $\times$ four attempts. The middle rows of each block share every cell and input with the thinking row; only `enable_thinking` differs. Registered primary: p2t – p2n; the others are secondary.

4 The loop without selection

Iteration 1. The base, thinking, tried once each of L1, L2 and L3 on the 48 training instances and produced 60 accepted proofs (21, 23 and 16 of 48), none at the first submission, at a median of about 100,000 output tokens per session. The proofs are the model’s own: with and without auxiliary lemmas, induction closed by `omega`, `norm_num`, `simp` or `linarith`. They were consolidated as they came (no selection, no editing, no ordering) into *own*; the reference proofs of exactly those 60 cells gave *curm*. Table 4 (rows *own*, *curm*) and Table 5 give the reading. Without thinking, *own* solves L1 at 24/24 (base 4/24), 23 of them at the first submission, and 1/24 at L2–L3 combined; with thinking it solves L2 at 21/24 and L3 at 19/24 against 6/24 and 8/24 for the base with the same thinking (+0.62 and +0.46, $p = 0.016$), and is within noise of the six-form template adapter p2 with thinking (19/24, 18/24). Nothing moved at L4, L5 or E, where the model had never produced proofs. The matched template control shows what the template adds: without thinking, *curm* solves L3 at the first submission in 13/24 where *own* solves 1/24; with thinking, *own* reaches *curm* spending roughly four to twenty times the tokens (medians 11,700 and 20,000 against 3,200 and 900), never at the first submission.

Iteration 2. The adapter *own*, thinking, tried once each of L4, L5 and E on the training instances and produced 75 accepted proofs: E 38/48, L5 35/48, L4 2/48. Consolidated cumulatively with the 60 of iteration 1 (135 examples), it gives *own2*; the 135 reference counterparts give *curm2*. *Own2* solves E in 24/24 and L5 in 16/24 without thinking, at the first submission, at a median of about 540 tokens per session, against 0/24 and 1/24 for the base without thinking and 12/24 and 10/24 for the base with 131k tokens of thinking (+1.00, $p = 0.016$; +0.62, $p = 0.031$). This lineage was never shown a template of E or L5. The base produced proofs of L1–L3; *own*, trained on them, produced proofs of E and L5; *own2*, trained on those as well, solves E and L5 at the first submission. With thinking, *own2* reaches 22/24 at both E and L5 against 12 and 10 for the thinking base, with $p = 0.062$ at both, five instances up and one tied, below the registered threshold and reported as such. Retention: no fall by the rule; descriptively, cumulative consolidation lowered the rate at one lower step (L1, 24/24 to 19/24, -0.21, $p = 0.062$) and left the others unchanged (Table 4). One cell of that table has no explanation we can offer: the template adapter of 135 cells (*curm2*) solves L3 without thinking in 13/24 where the six-form template adapter of 288 proofs (p2) solves 7/24, with one seed each.

arm	L1	L2	L3	L4	L5	E
<i>no thinking (16,384 tokens, T 0.7)</i>						
base	4/24	0/24	0/24	0/24	1/24	0/24
own: 60 own proofs of L1–L3 (iteration 1)	24/24	0/24	1/24	0/24	0/24	0/24
curm: the reference proofs of the same 60 cells	24/24	0/24	13/24	0/24	0/24	0/24
own2: 135 own proofs, cumulative (iteration 2)	19/24	0/24	1/24	0/24	16/24	24/24
curm2: the reference proofs of the same 135 cells	24/24	1/24	13/24	8/24	24/24	24/24
p2: reference proofs of all six steps (288)	24/24	1/24	7/24	24/24	24/24	24/24
<i>thinking (131,072 tokens, T 1.0)</i>						
base	–	6/24	8/24	9/24	10/24	12/24
own (iteration 1)	–	21/24	19/24	–	–	–
curm	–	17/24	18/24	–	–	–
own2 (iteration 2)	–	20/24	17/24	6/24	22/24	22/24
curm2	–	16/24	11/24	6/24	24/24	24/24
p2	–	19/24	18/24	–	–	–

Table 4: The loop, two iterations, six held-out instances $\times$ four attempts. Own and curm are trained on the same 60 cells; own2 and curm2 on the same 135; p2 on 288 reference proofs of all six steps. The thinking rows of iteration 1 were run at L2 and L3 only.

contrast	step	rate difference
own – base (no thinking)	L1	+0.83 ($p = 0.016$)
ownt – baset	L2	+0.62 ($p = 0.016$)
ownt – baset	L3	+0.46 ($p = 0.016$)
ownt – p2t	L2	+0.08 ($p = 0.250$)
ownt – p2t	L3	+0.04 ($p = 0.500$)
curm – own (no thinking)	L3	+0.50 ($p = 0.062$)
own2 – base (no thinking)	L5	+0.62 ($p = 0.031$)
own2 – base (no thinking)	E	+1.00 ($p = 0.016$)
own2t – baset	L4	-0.12 ($p = 0.844$)
own2t – baset	L5	+0.50 ($p = 0.062$)
own2t – baset	E	+0.42 ($p = 0.062$)
curm2 – own2 (no thinking)	L3	+0.50 ($p = 0.031$)
curm2 – own2 (no thinking)	L4	+0.33 ($p = 0.250$)

Table 5: The contrasts of the two iterations (paired rate difference over six instances, one-sided sign-flip p). Registered primaries: iteration 1, own – base (no thinking) and ownt – baset; iteration 2, own2 – base at L5 and E, own2t – baset at L4, L5 and E; the rest are secondary.

Where the loop stops: L4 and the mixed form. L4 is where the loop stops: production yielded two proofs in 48 sessions, and own2 solves 0/24 without thinking and 6/24 with it (base with thinking 9/24). Own2 declares a function of the state in 24 of 24 first L4 submissions, under names such as `inv`, `q` or `state_inv`, but of the wrong kind: a linear form in 16 and a diagonal quadratic form ($x^2 - 2y^2 + z^2$ with offsets) in 8. The mixed term that L4 requires, a product of two different fields such as $(x - 4)(z - 3)$, appears in 0 of 24, and in no submission of any session. The template adapters write the mixed term in 20 of 20 and 21 of 21 first submissions; the thinking base in 14 of 22 (13 by the classifier of Section 7); own2 with thinking in 1 of 24 first submissions and in 11 of 24 sessions at some point (6 solved). Two further facts bear on this. First, the two own L4 proofs that did enter own2’s data were of the mixed form, and so were the two reference proofs of the same cells in curm2’s data; the two template examples suffice for curm2 to write the mixed form in every first submission, while the two own examples leave own2’s first submissions unchanged. In the template set the L4 form is one more instance of a skeleton the adapter learned from all 135 examples; in the self-produced set it is two proofs among 135 heterogeneous ones. Second, the prior is conditional on the world’s shape. By the taxonomy of Section 7 the L5 form $x^2 - kxy + y^2$ is also a mixed quadratic, own2’s data hold

28 such proofs from L5, and own2 writes that kind in 24 of 24 first submissions at L5 and in 0 of 24 at L4: it produced the mixed form in the two-field worlds of the Vieta map and did not carry it to the three-field worlds of L4 with shear moves, where its own two productions were too few. (A first reading of this node counted the name `measure`, the name of our reference proofs, and concluded that the loop’s adapters “never declare a measure”; the count above, of what is written, replaced it on 19 September, and the classifier of Section 7 reproduces it: 16 linear, 8 diagonal, 0 mixed.)

5 Out of the training distribution

The wall search: no candidate met the definition. To ask whether the loop passes a step the base never passes, four candidate families were built (W1–W4, Section 2) and screened with three arms, eight sessions each: the base thinking; the base thinking with the complete reference proof of a *sibling* instance of the same family shown as a worked example; and own2 (Table 6). Three candidates fell at the screen. W1 (base 0/8) went to a confirmation on six new instances with eight attempts each: the base solved 6/48, on four of the six instances, and own2 30/48, on all six. The registered definition of a wall is a support statement at one budget (base 0 of 48 sessions at 131k tokens), not a statement of difficulty, so one solve is enough, and the 0/8 of the screen was compatible with a rate of about ten per cent. Own2’s rise (+0.50, $p = 0.016$) is recorded, but the registered crossing rule applies only to a confirmed wall. In 33 of 48 base sessions the model writes, at some submission, the quadratic form modulo p that the family needs, and in 45 some modular invariant (by the classifier of Section 7; the coarser count of sessions attempting anything modular, made at the time of the node, was 29): it has the right idea, and then fails to complete the Lean proof, ending at the 131k-token ceiling; the sibling proof unlocks it (19/24). A worked proof supplies the idea and the form together, so this test cannot tell which of the two the base was missing.

candidate	base	base + sibling proof	own2	class
W1 Pell form modulo a prime	0/8	6/8	7/8	wall_candidate
W2 plane lattice (trap)	4/8	7/8	3/8	not_a_wall
W3 hyperbolic form, four fields	2/8	8/8	7/8	not_a_wall
W4 large prime	3/8	7/8	7/8	not_a_wall
<i>confirmation of W1, six new instances</i>				
W1	6/48	19/24	30/48	not_a_wall_at_scale
curm2, the template adapter (descriptive, cut by the guard): 0/18; own2 – base +0.50 ($p = 0.016$)				

Table 6: The wall search: screen (eight sessions per candidate and arm) and the confirmation of the one candidate (six new instances; base and own2 $\times 8$, base with the sibling proof $\times 4$, template adapter $\times 4$, cut short at 18 sessions by the automatic removal deadline of the rented instance). Secondary contrasts only: the registered rule was the wall definition.

Generalisation against the template, on the same cells. The confirmation left a side finding: on W1 the template adapter curm2, trained on the same 135 cells as own2, solved 0/18 and never tried anything modular, while own2 solved 30/48. Node A8 then measured it: base, own2 and curm2, thinking, on six fresh instances of each of the four families, three attempts each, with two rules registered before the data (*narrowing*: curm2 falls against the base in at least two of four families; *generalisation*: own2 rises against the base in at least two). The node’s read, made with the template arm at 27 of 72 cells after three operational failures to complete it, gave `self_generalises_only`; a second read, declared as such before any new cell and made with the arm complete, gave the same verdict (Table 7). Own2 rises against the base in W1, W3 and W4 by the rule (52/72 in all, against 15/72; W2 +0.28, $p = 0.125$), at a

seventh to two fifths of the tokens by family (medians of 19,000 to 52,000 against 100,000 to 131,000). Curm2 does not narrow below the base: 16/72 against 15/72, with no family falling by the rule. Its deficit is against the loop’s adapter: by the rule in W1 and W3; in W2 (+0.22, $p = 0.062$) and W4 (+0.44, $p = 0.078$) short of the rule on p . Table 8 gives the same data as instance coverage: at three attempts the base solves at least one session on 0, 4, 4 and 3 of the six instances of W1–W4, the template adapter on 0, 6, 1 and 4, and own2 on all six of every family.

family	base	own2	curm2 (template)	own2 – base	curm2 – base	own2 – curm2
W1	0/18	13/18	0/18 (node read 0/4)	+0.72 ($p = 0.016$) $\uparrow$	+0.00 ($p = 1.000$)	+0.72 ($p = 0.016$) $\uparrow$
W2	6/18	11/18	7/18 (node read 1/6)	+0.28 ($p = 0.125$)	+0.06 ($p = 0.500$)	+0.22 ($p = 0.062$)
W3	5/18	14/18	3/18 (node read 1/9)	+0.50 ($p = 0.031$) $\uparrow$	-0.11 ($p = 0.375$)	+0.61 ($p = 0.031$) $\uparrow$
W4	4/18	14/18	6/18 (node read 3/8)	+0.56 ($p = 0.031$) $\uparrow$	+0.11 ($p = 0.375$)	+0.44 ($p = 0.078$)

Table 7: Generalisation against the template, four new families, six fresh instances each, three attempts, thinking. The template column is the complete arm of the second read, with the node read’s partial count in parentheses; base and own2 are the same in both reads. Registered rules: own2 – base rises in at least two families; curm2 – base falls in at least two. own2 – curm2 is secondary.

block	arm	attempts per instance	instances with at least one solve
W1 (A7 confirmation)	base	8	W1 4/6
W1 (A7 confirmation)	own2	8	W1 6/6
W1 (A7 confirmation)	base + sibling proof	4	W1 6/6
W1 (A7 confirmation)	curm2, template (cut short)	2–4	W1 0/6
W1–W4 (A8)	base	3	W1 0/6, W2 4/6, W3 4/6, W4 3/6
W1–W4 (A8)	own2	3	W1 6/6, W2 6/6, W3 6/6, W4 6/6
W1–W4 (A8)	curm2	3	W1 0/6, W2 6/6, W3 1/6, W4 4/6
W1–W4, fresh instances (A9)	own2	3	W1 6/6, W2 6/6, W3 6/6, W4 6/6
W1–W4, fresh instances (A9)	own3	3	W1 4/6, W2 6/6, W3 6/6, W4 6/6
W5 (A9)	base	3	W5 1/6
W5 (A9)	own2	3	W5 4/6
W5 (A9)	own3	3	W5 6/6
ladder, thinking (iteration 2)	base	4	L4 6/6, L5 6/6, E 5/6
ladder, thinking (iteration 2)	own2	4	L4 3/6, L5 6/6, E 6/6
ladder, thinking (iteration 2)	curm2	4	L4 2/6, L5 6/6, E 6/6

Table 8: Support at small budgets: instances with at least one accepted proof, at the attempts each block ran (descriptive). No block measured more than eight attempts per instance.

The template adapter’s first submissions are uniform within each family. They are the same kind of invariant in 18 of 18 sessions: the exact Pell form in W1 (never the modulus; 0/18 solved), the exact linear form in W2 (the trap; 8 sessions later repair it to a modulus and 7 solve), the bilinear form with offsets in W3 (the right kind; 3/18 solved), and the L3 measure (...) mod p in W4 (the right kind; 6/18). Where the obstruction is the combination of two trained skeletons (a quadratic form and a modulus in W1; a linear form and a modulus in W2), it attempts the combination in 8 of 36 sessions, the base in 35 of 36 and the loop’s adapter in 36 of 36; it never adds a modulus to a quadratic form. The two adapters differ in how often they combine skeletons and in which combinations they attempt: the template adapter writes its nearest trained skeleton and repairs only by appending a modulus to a linear form; the loop’s adapter starts from a wider set of forms and combines them (repair rates in Section 7).

6 The third iteration saturates

The loop was turned once more with what it had just produced: the 52 proofs own2 wrote on the wall families were added to its data (187 examples; the 24 instances of A8 became training instances) and consolidated into *own3*. Three questions were registered: does own3 *compose* (the primary: rise against own2, thinking, on W5, a new family that needs the form of W3 conserved only modulo a prime as in W1); does it *cheapen* (rise against own2 without thinking on at least two of W1–W4, on 24 fresh instances); does it *retain* (no fall against own2 on the six ladder steps without thinking, nor on W1–W4 with thinking). Table 9 gives the read: **saturates**. On W5 own3 solves 9/18 against 7/18 for own2 ($+0.11$, $p = 0.375$), with six instances and own2 already at 0.39, a test of little power. Without thinking, nothing moves on the new families (3/72 against 0/72, every family on the floor). There is no fall by the rule anywhere (descriptively, E 24/24 $\rightarrow$ 21/24). A plateau after two or three iterations is what the self-training literature reports [35, 47], and this result replicates it on a new substrate; what did change at the third iteration is described next.

block	own3	own2	base	own3 – own2	note
W5, thinking (primary)	9/18	7/18	1/18	$+0.11$ ($p = 0.375$)	own3 – base $+0.44$ ($p = 0.016$) $\uparrow$
W5, no thinking	0/18	0/18	–	$+0.00$ ($p = 1.000$)	
W1 new instances, no thinking	2/18	0/18	0/18	$+0.11$ ($p = 0.250$)	
W2 new instances, no thinking	1/18	0/18	0/18	$+0.06$ ($p = 0.500$)	
W3 new instances, no thinking	0/18	0/18	0/18	$+0.00$ ($p = 1.000$)	
W4 new instances, no thinking	0/18	0/18	2/18	$+0.00$ ($p = 1.000$)	
W1 new instances, thinking	8/18	9/18	–	-0.06 ($p = 0.500$)	
W2 new instances, thinking	15/18	13/18	–	$+0.11$ ($p = 0.312$)	
W3 new instances, thinking	14/18	11/18	–	$+0.17$ ($p = 0.250$)	
W4 new instances, thinking	12/18	14/18	–	-0.11 ($p = 0.391$)	
ladder L1, retention, no thinking	20/24	19/24 (16 Sept.)	–	$+0.04$ ($p = 0.500$)	
ladder L2, retention, no thinking	2/24	0/24 (16 Sept.)	–	$+0.08$ ($p = 0.500$)	
ladder L3, retention, no thinking	2/24	1/24 (16 Sept.)	–	$+0.04$ ($p = 0.500$)	
ladder L4, retention, no thinking	2/24	0/24 (16 Sept.)	–	$+0.08$ ($p = 0.500$)	
ladder L5, retention, no thinking	17/24	16/24 (16 Sept.)	–	$+0.04$ ($p = 0.500$)	
ladder E, retention, no thinking	21/24	24/24 (16 Sept.)	–	-0.12 ($p = 0.250$)	
W5 solved by the intended route (quadratic form modulo p), post hoc: base 1/18, own2 4/18, own3 8/18					

Table 9: The third iteration, all blocks complete (594 sessions, none lost). Own3 and own2 are compared on identical cells; the ladder comparator is own2’s run of 16 September on the same cells. Registered primary: own3 – own2 on W5 with thinking; own3 – base on W5 is secondary.

Four descriptive observations accompany the verdict. First, the loop’s adapters solve a family the base barely does: W5 is hard for the base (1/18, one instance, 13 sessions at the token ceiling); own3 and own2 solve 9/18 and 7/18 on six and four instances, and own3 against the base is $+0.44$ ($p = 0.016$), a secondary contrast. Second, the third iteration raised the rate of the needed kind at the first submission without raising the solve rate. Thinking, own3 writes the needed kind of invariant at the first submission in 13/18, 9/18 and 17/18 sessions of W1–W3 against 4/18, 0/18 and 5/18 for own2, and the totals solved are 49/72 and 47/72; own3 writes at the first submission the kind own2 reached only after a rejection (at W3 this cuts the tokens to a quarter, medians 17,000 against 61,000). Without thinking, the first submission moves in the same way but the proof is not completed: own3 writes the needed kind in 14/18 (W1) and 18/18 (W3) first submissions and solves 2 and 0. Third, without thinking the loop’s adapter behaves like the template one. On W5 it writes the skeleton of W3, the exact bilinear form, in 18 of 18 first submissions and never the composition; with thinking, the same weights attempt the composition in 14 of 18 sessions (own2 in 7, the base in 12). In these sessions the combination of two forms appeared only in the thinking configuration (14 of 18 sessions against 0 of 18 without thinking), while each individual form appeared in both. Fourth, the

same movement reaches the ladder: on the held-out L4 instances without thinking, own3 writes the mixed kind in 21 of 24 first submissions, where own2 wrote it in 0 of 24, and in 13 of the 24 the function it declares is the world’s reference measure exactly, offsets included (post hoc, 25 September, by polynomial comparison with the hidden reference measure); it solves 2 of 24 (own2 0 of 24), both among those 13, and the other eleven sessions with the exact measure end without an accepted proof. The 52 proofs added at the third iteration include the 14 W3 proofs, whose invariant is a mixed quadratic form in four fields with offsets; the kind carried to the three-field worlds of L4, and the execution did not follow.

A defect of the instrument. Reading the accepted W5 proofs exposed two shortcut families the certificate did not test: order invariants (W5 has no inverse moves, so a region such as $x \geq a \wedge y \geq b$ can be closed under the moves with the target outside it) and the modular orbit of a pair of linear forms $(x + z, y + w)$. Four of the seventeen accepted W5 proofs used one of them (own2 three, own3 one); by the intended route the counts are base 1, own2 4, own3 8 of 18, and the post-hoc contrast $\text{own3} - \text{own2}$ is $+0.22$ ($p = 0.11$), which would not have passed the rule either. The second family also occurs in W1: 13 of its 100 accepted proofs close a set of residues of (x, y) modulo p without the quadratic form, including three of the base’s six in the A7 confirmation (by the intended route 3 of 48, on two of six instances), six of own2’s 37 in A7, two of the sibling arm’s 19, one of own2’s 13 in A8 and one of own3’s 8 in A9. No verdict changes: the verifier accepted these proofs, so they count under the registered rule, and the reads stand as registered. No W5 proof entered a training set, since own3 was trained before A9; own3’s data do contain shortcut proofs of other families (one W1 proof from A8 and seventeen ladder proofs at L2, E and L5 by an invariant of another kind than the intended one, Section 10), a risk that consolidation without selection carries by construction [33]. A further family would have needed inverse moves, or a certificate testing order invariants and residue sets of pairs; none was built, as the line of work closed on 25 September 2026.

7 What is tried, and what is executed

Several of the results above turned on the first submission. Node A6 measured it directly: for every observed prover session of the base and its adapters (3,660 sessions in 42 conditions), a heuristic classifier over the Lean source the verifier received labels the kind of invariant the first submission declares (*linear*, *linear modulo*, *diagonal quadratic*, *mixed quadratic*, *quadratic modulo*, *none*; Appendix B). The classifier was validated by labelling sessions by reading before seeing its label: 57 of 60 in a first stratified sample; after closing the two gaps that sample exposed (the ZMOD notation; invariance stated as a relation between two states), 40 of 40 in a second, fresh sample, and 99 of the 100 overall with the corrected version. Each step has a needed kind fixed by its generator. For a condition and a step, π is the rate of the needed kind at the first submission, θ_{right} the solve rate of the sessions that started with it, θ_{wrong} that of the others; intervals are a Bayesian bootstrap over instances (Dirichlet weights, 4,000 replicates, Jeffreys smoothing), paired when two conditions share their instances. Two cautions apply to these quantities. Conditioning on the first kind conditions on a post-treatment variable, so θ is a descriptive quantity and cannot be read as a causal effect. And where every session of every instance has the kind, the smoothing floor prints as 0.98 or 0.02 with a degenerate interval, and a contrast between two such cells clamps at about seven nats; the tables print those cells as saturated rather than as estimates. Table 10 gives the contrasts and Appendix D the values per condition.

Four readings, each registered as a question with a prediction before the tables were run. (Q1) Template consolidation moves π to the ceiling at every step and moves θ only where the form has no parameters to infer: at L2 θ does not move; at L3 it rose 2.9 nats, against the

contrast	step	$\Delta \logit \pi$	$\Delta \logit \theta_{\text{right}}$	$\Delta \logit \text{rate}$
p2 – base, no thinking (replication)	L1	0 (sat.)	+5.37 [+5.04, +6.08]	+5.37 [+5.04, +6.08]
p2 – base, no thinking (replication)	L2	+3.52 [+2.73, +4.50]	+0.43 [-0.62, +1.53]	+0.96 [+0.12, +1.95]
p2 – base, no thinking (replication)	L3	+1.49 [+0.23, +2.75]	+2.92 [+1.95, +3.74]	+3.00 [+2.14, +3.77]
p2 – base, no thinking (replication)	L4	>+7 (sat.)	–	>+7 (sat.)
p2 – base, no thinking (replication)	L5	+4.78 [+4.43, +4.93]	+5.53 [+4.15, +6.50]	+6.79 [+5.79, +7.67]
p2 – base, no thinking (replication)	E	+5.16 [+4.63, +5.90]	+6.29 [+5.67, +6.67]	>+7 (sat.)
own2 – base, walls, thinking	W1	-1.91 [-2.79, -1.46]	–	+4.46 [+4.28, +5.03]
own2 – base, walls, thinking	W2	-2.69 [-3.75, -1.93]	+1.89 [+0.99, +3.01]	+1.09 [+0.40, +2.03]
own2 – base, walls, thinking	W3	+0.54 [-0.62, +1.13]	+2.31 [+0.56, +4.06]	+2.10 [+1.17, +3.27]
own2 – base, walls, thinking	W4	-3.03 [-4.15, -1.99]	+1.96 [+0.78, +3.36]	+2.42 [+1.37, +3.56]
curm2 – base, walls, thinking	W1	-3.62 [-4.01, -3.21]	–	0 (sat.)
curm2 – base, walls, thinking	W2	-5.10 [-5.98, -4.55]	–	+0.23 [-0.38, +0.97]
curm2 – base, walls, thinking	W3	+3.66 [+2.86, +4.11]	-1.46 [-3.08, -0.25]	-0.81 [-2.45, +0.65]
curm2 – base, walls, thinking	W4	+0.98 [+0.13, +2.03]	+0.50 [-0.66, +1.55]	+0.57 [-0.57, +1.61]
own2 – curm2, walls, thinking	W1	+1.68 [+0.69, +2.44]	–	+4.46 [+4.28, +4.98]
own2 – curm2, walls, thinking	W2	+2.48 [+1.77, +2.84]	–	+0.88 [+0.45, +1.24]
own2 – curm2, walls, thinking	W3	-3.12 [-3.54, -2.97]	+3.78 [+2.25, +5.53]	+2.94 [+1.18, +4.73]
own2 – curm2, walls, thinking	W4	-4.03 [-4.67, -3.50]	+1.53 [-0.12, +3.42]	+1.84 [+0.62, +3.45]
own2 – curm2, L4, no thinking	L4	-5.68 [-7.36, -4.11]	–	-3.15 [-4.51, -1.59]
own2 – base, L4, thinking	L4	-3.13 [-4.19, -1.68]	–	-0.55 [-1.92, +0.44]
p2t – p2n (thinking given the form)	L2	-0.96 [-1.97, -0.11]	+3.72 [+2.75, +5.07]	+3.52 [+2.66, +4.67]
p2t – p2n (thinking given the form)	L3	-1.66 [-2.38, -0.69]	+2.72 [+2.26, +3.30]	+2.37 [+1.81, +3.04]
baset – basen (thinking without the form)	L2	+1.03 [+0.55, +1.57]	+1.55 [+0.42, +2.60]	+1.25 [+0.39, +2.52]
baset – basen (thinking without the form)	L3	-0.85 [-1.69, +0.05]	+3.09 [+2.61, +3.62]	+3.21 [+2.94, +3.58]

Table 10: Contrasts in nats (difference of logits), median and 90% interval of the Bayesian bootstrap; rows share their instances and are paired. “sat.”: both cells at the smoothing floor or ceiling, a clamp rather than a measured difference. Descriptive; no rule.

prediction, to 0.29, still far from one. (Q2) In three of the four wall families the loop’s adapter has a lower π than the base at the first submission (0.12, 0.24 and 0.39 at W1, W2 and W4 against 0.50, 0.82 and 0.93; at W3, 0.62 against 0.49). It starts with the simplest form of its repertoire, is rejected, and repairs: of the sessions that start with the wrong kind, 10 of 16 in W1, 14 of 14 in W2 and 9 of 11 in W4 reach the needed one, and 24 of them solve. The base almost never repairs (0 of 9 in W1) and, when it starts with the needed kind, rarely completes the proof (W1 0/9, W2 4/15, W4 4/17). The template adapter has π at the ceiling where a trained skeleton serves and at the floor where two must be combined, repairs only in W2 (8 of 18 sessions, 7 solved) and executes the needed kind poorly (θ_{right} 0.15 at W3 against 0.89 for the loop). Out of distribution, the loop’s adapter improved in execution and in repair after a rejection rather than in its initial guess. (Q3) At L4 the second iteration’s adapter fails at the first guess: π is 0.02 without thinking and 0.05 with it, against 0.85 for the template adapter and 0.54 for the thinking base; of the 23 thinking sessions that start with another kind, 10 reach the mixed one later and 5 of those solve, and the 13 that never reach it solve none. After the third iteration, with fourteen four-field mixed proofs of W3 in the data, π at L4 without thinking is 0.86 (in 13 of 24 sessions the exact reference measure, counted post hoc on 25 September) and the solve rate 0.08: the prior moved to the needed kind and the execution did not follow. (Q4) With the form in the weights, thinking moves θ (+3.7 and +2.7 nats at L2 and L3) and lowers π slightly; without the form it moves θ too, and π only at L2.

Under thinking, the first submission is what the verifier received after the model reasoned, and it was often extracted from the reasoning itself: on the walls, at least one submission was extracted from the reasoning in 10 of 72 base sessions, 26 of 72 own2 sessions and 68 of 72 template-adapter sessions. So π with thinking is the prior after thinking, and the gap between the no-thinking and thinking rows of the same weights measures what reasoning does to the

first guess.

In these data the kind of invariant tried first tracks the forms in the training data, conditionally on the shape of the world: the second iteration’s adapter writes the mixed form where it produced it (L5) and not at L4, the third writes it at L4 once four-field mixed proofs are in its data, and the template adapter writes its trained skeletons and rarely combines two. Execution and repair improved with thinking and with practice on verified proofs, and those are the quantities that rose on the new families.

8 Discussion

What the loop did, stated within its measurements. Every step family that an adapter solved, the base also solved at least once at up to eight attempts and 131k tokens of thinking: the one attempt to construct a step the base never passes failed because the base passed every candidate at least once, and the two rungs the loop climbed were rungs where the thinking base already solved four or five of ten (three or four by the intended route, Section 10). At the level of instances the picture differs by block (Table 8). On the one family measured at eight attempts (W1), the base reached four of six instances and own2 all six; on the four new families at three attempts, own2 covered every instance where the base covered eleven of twenty-four, and solved 3.5 times as many sessions (five times on W1 at eight attempts, seven times on W5, nine for own3); on the ladder with thinking own2 solved 1.6 times as many sessions as the base at L4, L5 and E (curm2 1.7 times), with L4 the exception in the other direction (own2 6/24 against the base’s 9/24, and three of six instances against six). The third iteration, trained on the adapter’s own proofs of the new families, changed the first guess without raising the solve rate. Whether the base’s pass@ k at larger k would reach every instance the adapters reach is not measured here; a pass@ k curve to k of 32 or 64 on two families would cost one night of GPU and would settle it; it was not run, and the line of work closed on 25 September 2026. The mechanism that fits every reading is sharpening [14]: consolidation on self-verified samples raises the probability of what the model can already produce, which is the weight-side reading of the report that reinforcement learning with verifiable rewards sharpens the base without extending its pass@ k frontier [45], a report itself disputed on the size of k and the length of training [4, 16, 21, 38]. The loop is also a form of distilling reasoning into the no-thinking mode [43]: iteration 1 takes thinking-mode solutions at 100,000 tokens and trains the no-thinking mode to emit them, on a model whose no-thinking mode is already a distillate of its thinking mode [30]. The result of the second paper of this series, that consolidation moved the mass of proposals and not the frontier [24], holds here as well; this time no step fell by the rule at any iteration, and the collapse that self-generated data can cause [10, 34] did not appear in three iterations. The paper claims nothing beyond them.

What is new against expert iteration. The loop is expert iteration [2, 28, 29] in the STaR/ReST family [12, 35, 44, 47], whose success in Lean is established at scale [11, 20, 32, 39, 40] and whose self-play versions remove the human from statement generation as well [7, 18, 19, 48]; rejection-sampling fine-tuning on verified positives is competitive with policy-gradient training [41], and updating the weights on the problem at test time is the same lever applied within a single task [37, 46]. Against that line the present loop is small and still has a human-designed generator, ladder and curriculum. Its contribution is the control and the instruments: the same cells with a template counterpart, a ladder whose steps and needed forms are known, held-out instances that differ only in constants, and a read of forms as well as of rates. The central contrast is own against curm on identical cells. Both adapters are supervised fine-tunes that memorised their examples. The one trained on the model’s own heterogeneous verified proofs rose on three of the four new families; the one trained on uniform template proofs wrote

its nearest trained skeleton and rose on none. That contrast confounds three things the design did not separate: the origin of the data (on-policy against off-policy), its diversity, and the uniformity of the template. Each has a literature that predicts the direction observed. Self-generated data closes the distribution gap that causes forgetting after fine-tuning [22, 42], and on-policy data is the standard remedy for the exposure mismatch of imitation [1]; coverage from a weaker but more diverse source beats a stronger uniform one [3]; fine-tuning on few examples teaches format rather than knowledge [49]; and supervised fine-tuning memorises where policy learning generalises [5], with the variable here being the data’s origin and diversity rather than the algorithm. One adapter would separate them: the model’s own proofs filtered to a single skeleton, or a diverse off-policy set of verified proofs of the same cells; neither has been run.

Repair under verifier feedback. Out of distribution the loop’s adapters gain by repairing after a rejection; without thinking the same adapter repeats its first form, and the base, given the same rejection, rarely repairs (0 of 9 in W1). Self-correction without external feedback is weak [15]; repair from a proof assistant’s errors is a known route [8], and self-correction can be trained [17]. Here the feedback is exact, and the loop trained the capacity to use it. It is also the place where the session carries the least information: between submissions only the submitted file and the errors return, and 50,000 to 100,000 tokens of reasoning are discarded. A channel between submissions, first in words and then, if words help, a trained one on the model’s own cache [9], was drafted as the next instrument, in a design note that was never pre-registered, and was not run.

The prior over forms. Reading the first submission as a draw from a distribution over forms, and success as conditional on the form, showed two things that rates alone did not: the loop’s adapter starts with the needed kind less often than the base and solves more because it repairs, and every consolidation moved π toward forms in its training data. The open problem of the series is unchanged: nothing measured so far moves the first guess toward a form the model has never produced unless that form is shown to it.

9 Limitations

One model (a served 27B), one training recipe, one generator of formal worlds written by us, six held-out instances per step or family, three or four attempts per instance, one seed per adapter (the replication changed data and seed together; the loop’s adapters are unreplicated). Each iteration retrained from scratch on the cumulative set; continuing from the previous adapter was not tried, and “saturates” is measured under that choice. The rule has the power of six paired instances, is read next to ceilings and floors that are reported and not corrected, and its per-contrast level is uncorrected for multiplicity; the registered primaries and conjunctions are the protection. The thinking comparators of the two iterations and the ladder comparator of the third were served on other days (same weights, sampling and cells); a day effect would bias the retention readings in either direction and is unmeasured. Support beyond eight attempts per instance was not measured, so “within the base’s support” is a statement at that budget. The worlds were generated in September 2026, after the model’s release, so their instances cannot be in its training data; their proof skeletons resemble textbook invariant arguments, and “never produced” means never produced in these runs. The template arm of A8 was completed in a second read declared as such. The wall certificate missed two shortcut families, in W5 and, for one of them, in W1 (Section 6); the ladder had no certificate, and some of its worlds admit shortcut invariants (Section 10). The classifier of forms is heuristic, validated on samples of 60 and 40, does not distinguish the two-field mixed form of L5 from the three-field one of L4, and θ conditions on a post-treatment variable; the Bayesian bootstrap intervals are descriptive

and were not used as tests. Costs and failures of operation are recorded in Appendix A; none affected an observation that was read.

10 Corrections after freezing

Draft 0.4 was frozen on 25 September 2026. Notes (i) and (ii) were added on 26–27 September 2026 and corrected on 5 October 2026; the corrections listed under (iii) were made on 5 October 2026 for this version, after an audit of the draft against the records. Counts marked post hoc belong to no registered read; they were made with the classifier of Section 7 applied to the accepted proofs and checked by reading them; the scripts of these checks are in the repository under `scripts/paper6_checks/` and in the evidence deposit. No note or correction changes a registered verdict.

(i) Shortcut invariants on the ladder. The ladder had no shortcut certificate. A later check found that 75 of the generator’s 132 E worlds, 66 of the 108 used here and five of the six held-out instances of each held-out set, admit the parity of y as a separating invariant, and that some L5 worlds admit residue invariants of the same kind. At E and L5, in the held-out readings only the thinking base used them (3 of its 12 accepted E proofs, 2 of its 10 at L5); by the intended route it solves E in 9/24 and L5 in 8/24, and $\text{own2t} - \text{baset}$ would be +0.54 and +0.58 ($p = 0.016$, post hoc); the registered read stands. Of iteration 2’s production, 5 of 38 E proofs and 6 of 35 L5 proofs use such invariants (a seventh L5 proof parametrises the orbit), and they entered own2 ’s and own3 ’s data. At L2 every world also admits a separating linear form modulo a small integer (the exact form reduced modulo any integer that does not divide its difference between start and target, and in some worlds a simpler one, such as the parity of one field). At least 15 of the 108 accepted held-out L2 proofs use one instead of the exact form (the thinking base 3 of its 6, own with thinking 6 of its 21), and so do 6 of iteration 1’s 23 L2 proofs, which entered the data of own , own2 and own3 (post hoc, 5 October 2026). By the exact form, $\text{ownt} - \text{baset}$ at L2 would be +0.50 and $\text{p2t} - \text{p2n}$ +0.67 (both $p = 0.016$, post hoc), and own with thinking would solve L2 in 15/24 against 18/24 for p2 with thinking (Sections 3 and 4). The rates at L2, L5 and E are therefore rates of a valid proof, not of the intended invariant; at these steps the needed kind is not unique, and π counts a modular first submission as the wrong kind there.

(ii) The range of the constants. In a separate line of work on small models trained from scratch on the same kind of invariant worlds (repository, `docs/E3_PRECO_DE_UM_OPERADOR_RESULTADOS_V1.md`, `docs/E4_PISTAS_NOS_PESOS_RESULTADOS_V1.md` and `docs/E5_LISTA_OU_REGUA_RESULTADOS_V1.md`), one form of the dissociation observed here, the right kind of invariant written with wrong constants (Section 7), reappeared exactly, and there its cause was the *range* of the constants. A model that has seen the modular operator with moduli 3, 5 and 7 selects it with certainty on a world whose modulus is 11 to 19 and never writes the modulus; diversity of training families buys the form, not the range, and consolidating verified solutions with new moduli adds them as entries without producing an unseen one above chance (one held-out world of 60 at modulus 17 per seed; none at 31 or 37). The range does not explain the cases here: the held-out constants of L2 and L3 all lie in the range of the training instances (every held-out α , β and n occurs among them), and at L4 own3 wrote the exact reference measure in 13 of 24 first submissions and failed to complete the proof in 11 of them (Section 6).

(iii) Other corrections in this version.

- Section 6: the residue shortcut in W1 (13 of 100 accepted proofs) was added; the statement

that a W5 shortcut proof had entered own3’s data was false (own3 was trained before A9) and was replaced; the plan for a further family was replaced by the closure of the line of work. Section 2 and the Limitations were adjusted to match.

- Section 5: of the 45 base sessions of the W1 confirmation that wrote a modular invariant, 33 wrote the quadratic form modulo p ; the earlier text called all 45 the right idea.
- Section 4: the thinking base writes the mixed term in 13 of 22 first L4 submissions by the classifier of Section 7 (14 by the count of the node).
- Sections 6 and 7: own3’s exact reference measure at L4 (13 of 24) is marked post hoc; it was counted on 25 September, after the reads.
- Section 8: the multiple on W5 is seven for own2 (nine was own3’s); the multiple of 1.6 on the ladder with thinking is own2’s at L4, L5 and E (curm2’s is 1.7); the rungs the loop climbed are given also by the intended route; the three sentences on further instruments now say that none was run, that the line of work closed on 25 September 2026, and that the design note for a channel between submissions was never pre-registered.
- Section 3: the final training losses of the template adapters were at most 3.4×10^{-3} , not 10^{-3} .
- Section 2: the training seed of the replication was 2026091402, not 2026091302.
- Section 1: A6’s scripts were committed with its first read and rerun twice, which the earlier text, saying that every node was read once by a script written before its data, did not allow for; Appendix A gives the commit.
- Front matter, Availability and Acknowledgements: the version line; the two review rounds described as review by AI reviewer agents, not peer review; the models that assisted; the deposit; the adapter hashes (Appendix E). Figure 1 is now cited in the text, a broken cross-reference in note (ii) was repaired, and the appendix table of π was resized to fit its page.

11 Availability

Every node’s dated pre-registration and amendments, single-read script, read, diagnostics and complete evidence archive (raw session traces, verifier reports, plans, adapter manifests and weight hashes, GPU and collection logs, with a SHA-256 manifest) are in the project’s private working repository under `docs/` (A2_*, A6_*, A7_*, A8_*, A9_* and the *_evidence_v1 directories); the pre-registration commits are listed in Appendix A, and the tables and figures of this paper are generated from the read files by `scripts/paper6_tables.py` and `scripts/paper6_figures.py`. The generators of the ladder and wall families (`ladder_v1`, `ladder_walls_v1`), the prover session, the consolidation script (`a2c_train.py`) and the analysis scripts are in the same repository. Its public mirror, <https://github.com/RobertoOno/interrupting-the-loop>, stops on 10 September 2026 and does not contain this report’s material; all of it (the archives, the result documents, the code named above and the post hoc checks) is deposited at Zenodo, [doi:10.5281/zenodo.23163299](https://doi.org/10.5281/zenodo.23163299). The eleven adapters (about 3.5 GB) are not deposited; the SHA-256 hashes of their weight files are listed in Appendix E and recorded in the archives.

Two properties of this record bear on how the paper should be read. Every confirmatory claim is tied to a pre-registration dated before its data and to a read produced by a script committed before the data existed. And every number in the tables and figures is generated from the read files by scripts in the repository, so that a reader can regenerate them and follow each back to the raw sessions and verifier reports.

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The author designed the program, took every decision recorded in the notebook, and read and interpreted the results. An AI assistant (Claude, Anthropic: model Fable 5.1 between 13 and 27 September 2026, and model Opus 5 from 15 to 18 September, for the read of iteration 2, A7, and the pre-registration and operation of A8; this version’s corrections were made with Opus 5.5 on 5 October 2026) wrote and ran the code, operated the data collection under the author’s direction, and drafted this manuscript from the notebook and the read files; the pre-registrations, with their rules and verdict vocabularies, were drafted by the assistant from the author’s questions and committed with the author’s approval before the corresponding sessions ran. The two rounds of review were done by two AI reviewer agents (instances of Fable 5.1, each with its own brief and without seeing the other’s report), not by peer reviewers; they read the first drafts, and their reports and the author’s responses are archived in the repository (`docs/reviews/`, `docs/RESPOSTA_PARECER_PAPER6.md`). The author is responsible for the content.

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A Operations, amendments and pre-registration commits

Each node ran as one automated sequence on one rented GPU (install, serve, collect with a resume loop for transport failures, read, archive, commit, remove the instance) with an independent removal deadline and, from 19 September, a read-only watcher polling the server’s own request counters. Amendments recorded before the affected cells, by node: A2 replication, amendments 1–8 (driver version; thinking budget; concurrent collections; server restart; submission extracted from the reasoning when the content is empty, with the earlier cells archived unread; proxy cuts as infrastructure failures; removal deadline; the same-configuration arms).

Iteration 2: two observations before any test cell, the second correcting the first. A7: amendment 1 (confirmation run immediately after the screen), amendment 2 (template arm restarted after a machine reboot left the instance uncontrolled for two hours), a recorded transport-cut selection bias (retried cells are shorter sessions; the conclusion survives on the never-cut subset). A8: amendments 1–3 (three failures to complete the template arm, the third leaving an instance idle for five hours), amendment 4 (the second read declared). A9: amendment 1 (twelve client sessions instead of sixteen, because requests beyond the server’s cache waited unanswered until the proxy cut them), amendment 2 (TCP keepalive and a 600s cap on the silence between stream chunks, after twelve sessions lost their connection at the same second); after it, zero transport failures in 594 sessions. Total GPU of the program: about 120 hours of H200, US\$544 at list price, the sum of Table 1 (A9 includes the training of own3; the iterations include the training of their adapters).

node	pre-registration (commit, local time)	analysis script (commit)
A2	eba27556, 13 Sept. 17:17	analysis_a2_consolidation.py (32dfa45d, 13 Sept. 17:16)
A2 replication	7151aaa2, 14 Sept. 03:13	the same script, version 2
Loop, iteration 1	b48cd7d7, 14 Sept. 22:01	analysis_a2_own.py (same commit)
Loop, iteration 2	58edb12f, 15 Sept. 15:39	analysis_a2_own2.py (same commit)
A7	b7811841, 16 Sept. 21:18	analysis_a7_screen.py (same commit)
A8	f9a80467, 17 Sept. 20:59	analysis_a8_narrow.py (same commit)
A9	ca0b8e1c, 19 Sept. 08:16	analysis_a9.py (same commit)
A6	f8381cce, 19 Sept. 08:19	a6_forms.py, a6_model.py (abc55f15, 19 Sept. 08:28, with the first read)

Table 11: First commit of each node’s pre-registration document and of its analysis script, in the project repository (times in the laboratory’s local zone).

B The classifier of forms

The invariant’s text is, in order, the right-hand sides of the functions of the state the submission defines (any name); if none, the statements of its declarations that speak of invariance (mentioning a move, **Step** or **Reaches**, or relating two states); if none, the whole body. A product whose two factors both contain a state field is quadratic (mixed if the fields differ, diagonal if they coincide or one factor is raised to a power); a modular or divisibility marker in that text ($\%$, **ZMod**, **ZMOD**, **ModEq**, $\equiv$, $|$) makes it modular. Needed kinds: L1, L2 linear; L3, W2, W4 linear modulo; L4, L5, W3 mixed quadratic; E diagonal quadratic; W1, W5 quadratic modulo. The needed kind is not unique at L2, where every world also admits a separating linear form modulo a small integer, at E, where 66 of the 108 worlds admit the parity of y , and in some L5 worlds, which admit residue invariants (Section 10); there π counts such a first submission as the wrong kind. The classifier does not distinguish the two-field mixed form of L5 from the three-field one of L4. On the two validation samples together (100 sessions labelled by reading before seeing the automatic label), the corrected classifier agrees on 99; per class, with the hand label as reference: linear 30 of 30 (precision 0.97), linear modulo 24 of 24, diagonal quadratic 14 of 14, mixed quadratic 15 of 15, quadratic modulo 3 of 3, none 13 of 14 (a proof by parametrisation of a Fibonacci sequence, outside the taxonomy, read as linear). The two samples, their hand labels and the agreement files are archived with the node.

C The prover prompt

The user turn of a session is the following text, with the world’s Lean definitions and the target statement filled in (no system prompt; no example in the arms of this paper, except the sibling-proof arm of the wall screen, whose worked example is prefixed with “Below is a worked example: a separate formal problem and its complete proof, supplied for reference. It may or may not be

useful for the target that follows it.”):

Complete the following Lean 4 code:

```
```lean4
import Mathlib
import Aesop
set_option maxHeartbeats 400000
open BigOperators Real Nat Topology Rat
[the world's definitions: the state, the four moves, Step, Reaches, start]
theorem evaluation_solution : [the target statement] := by
```
```

Before producing the Lean 4 code to formally prove the given theorem, provide a detailed proof plan outlining the main proof steps and strategies.

After a rejected submission the next user turn is “The proof was rejected by Lean 4:” followed by the last 6,000 characters of the checker’s errors and “Fix the proof and provide the complete corrected Lean 4 code in a single ```lean4 code block, keeping the theorem statement unchanged.”

D The prior over forms per condition

| condition | step | n | solved | π | θ_{right} | θ_{wrong} |
|--|------|-----|--------|--------------------|-------------------------|-------------------------|
| <i>A2 replication, no thinking</i> | | | | | | |
| base | L1 | 24 | 4 | ≥ 0.98 (sat.) | 0.19 [0.10, 0.24] | – |
| base | L2 | 24 | 0 | 0.59 [0.35, 0.77] | 0.03 [0.03, 0.05] | 0.05 [0.03, 0.08] |
| base | L3 | 24 | 0 | 0.92 [0.77, 0.97] | 0.02 [0.02, 0.03] | 0.20 [0.08, 0.45] |
| base | L4 | 24 | 0 | ≤ 0.02 (sat.) | – | ≤ 0.02 (sat.) |
| base | L5 | 24 | 1 | 0.29 [0.26, 0.37] | 0.16 [0.07, 0.42] | 0.03 [0.03, 0.03] |
| base | E | 24 | 0 | 0.22 [0.12, 0.33] | 0.08 [0.06, 0.15] | 0.03 [0.02, 0.03] |
| p2 | L1 | 24 | 24 | ≥ 0.98 (sat.) | ≥ 0.98 (sat.) | – |
| p2 | L2 | 24 | 1 | ≥ 0.98 (sat.) | 0.05 [0.02, 0.13] | – |
| p2 | L3 | 24 | 7 | ≥ 0.98 (sat.) | 0.29 [0.15, 0.47] | – |
| p2 | L4 | 24 | 24 | ≥ 0.98 (sat.) | ≥ 0.98 (sat.) | – |
| p2 | L5 | 24 | 24 | ≥ 0.98 (sat.) | ≥ 0.98 (sat.) | – |
| p2 | E | 24 | 24 | ≥ 0.98 (sat.) | ≥ 0.98 (sat.) | – |
| <i>thinking (E8)</i> | | | | | | |
| basen | L2 | 24 | 2 | 0.47 [0.25, 0.64] | 0.04 [0.03, 0.07] | 0.15 [0.05, 0.34] |
| basen | L3 | 24 | 0 | 0.86 [0.79, 0.94] | 0.02 [0.02, 0.03] | 0.13 [0.09, 0.24] |
| baset | L2 | 24 | 6 | 0.70 [0.60, 0.81] | 0.18 [0.07, 0.35] | 0.44 [0.21, 0.68] |
| baset | L3 | 24 | 8 | 0.74 [0.65, 0.83] | 0.34 [0.25, 0.45] | 0.35 [0.16, 0.57] |
| p2n | L2 | 24 | 2 | ≥ 0.98 (sat.) | 0.10 [0.04, 0.18] | – |
| p2n | L3 | 24 | 5 | ≥ 0.98 (sat.) | 0.22 [0.12, 0.33] | – |
| p2t | L2 | 24 | 19 | 0.95 [0.87, 0.98] | 0.81 [0.76, 0.89] | 0.28 [0.14, 0.47] |
| p2t | L3 | 24 | 18 | 0.91 [0.82, 0.96] | 0.80 [0.73, 0.89] | 0.18 [0.10, 0.34] |
| <i>iteration 2, L4</i> | | | | | | |
| own2 (no thinking) | L4 | 24 | 0 | ≤ 0.02 (sat.) | – | ≤ 0.02 (sat.) |
| curm2, template (no thinking) | L4 | 24 | 8 | 0.85 [0.54, 0.97] | 0.39 [0.11, 0.74] | 0.12 [0.04, 0.41] |
| own2, thinking | L4 | 24 | 6 | 0.05 [0.02, 0.13] | 0.72 [0.53, 0.87] | 0.23 [0.09, 0.37] |
| base, thinking | L4 | 24 | 9 | 0.54 [0.44, 0.65] | 0.61 [0.48, 0.76] | 0.11 [0.05, 0.26] |
| <i>walls (A8, thinking; template arm complete)</i> | | | | | | |
| base | W1 | 18 | 0 | 0.50 [0.40, 0.60] | 0.05 [0.04, 0.06] | 0.05 [0.04, 0.06] |
| base | W2 | 18 | 6 | 0.82 [0.72, 0.92] | 0.28 [0.13, 0.45] | 0.64 [0.29, 0.86] |
| base | W3 | 18 | 5 | 0.49 [0.38, 0.68] | 0.45 [0.21, 0.67] | 0.13 [0.06, 0.30] |
| base | W4 | 18 | 4 | 0.93 [0.83, 0.97] | 0.24 [0.11, 0.41] | 0.28 [0.14, 0.47] |
| own2 | W1 | 18 | 13 | 0.12 [0.05, 0.23] | 0.50 [0.20, 0.80] | 0.73 [0.65, 0.84] |
| own2 | W2 | 18 | 11 | 0.24 [0.14, 0.32] | 0.73 [0.40, 0.89] | 0.57 [0.38, 0.75] |
| own2 | W3 | 18 | 14 | 0.62 [0.52, 0.66] | 0.89 [0.74, 0.95] | 0.57 [0.34, 0.78] |
| own2 | W4 | 18 | 14 | 0.39 [0.25, 0.53] | 0.69 [0.46, 0.88] | 0.80 [0.67, 0.91] |
| curm2 | W1 | 18 | 0 | ≤ 0.02 (sat.) | – | ≤ 0.02 (sat.) |
| curm2 | W2 | 18 | 7 | ≤ 0.02 (sat.) | – | 0.39 [0.34, 0.48] |
| curm2 | W3 | 18 | 3 | ≥ 0.98 (sat.) | 0.15 [0.04, 0.45] | – |
| curm2 | W4 | 18 | 6 | ≥ 0.98 (sat.) | 0.33 [0.17, 0.57] | – |
| <i>W5 (A9, thinking)</i> | | | | | | |
| base | W5 | 18 | 1 | 0.55 [0.42, 0.71] | 0.12 [0.05, 0.34] | 0.06 [0.04, 0.08] |
| own2 | W5 | 18 | 7 | 0.07 [0.03, 0.17] | 0.28 [0.14, 0.47] | 0.41 [0.22, 0.63] |
| own3 | W5 | 18 | 9 | 0.29 [0.16, 0.44] | 0.37 [0.13, 0.74] | 0.54 [0.42, 0.61] |
| <i>new wall instances (A9, thinking)</i> | | | | | | |
| own2 | W1 | 18 | 9 | 0.23 [0.08, 0.44] | 0.32 [0.12, 0.48] | 0.56 [0.39, 0.76] |
| own2 | W2 | 18 | 13 | ≤ 0.02 (sat.) | – | 0.70 [0.66, 0.80] |
| own2 | W3 | 18 | 11 | 0.27 [0.11, 0.52] | 0.75 [0.66, 0.87] | 0.54 [0.44, 0.61] |
| own2 | W4 | 18 | 14 | 0.52 [0.33, 0.62] | 0.87 [0.70, 0.94] | 0.67 [0.44, 0.87] |
| own3 | W1 | 18 | 8 | 0.73 [0.47, 0.88] | 0.39 [0.18, 0.62] | 0.60 [0.33, 0.75] |
| own3 | W2 | 18 | 15 | 0.51 [0.33, 0.63] | 0.87 [0.70, 0.94] | 0.76 [0.54, 0.90] |
| own3 | W3 | 18 | 14 | 0.93 [0.83, 0.97] | 0.75 [0.59, 0.88] | 0.72 [0.53, 0.86] |
| own3 | W4 | 18 | 12 | 0.50 [0.25, 0.74] | 0.75 [0.51, 0.90] | 0.54 [0.35, 0.75] |

Table 12: The prior over forms (π : the needed kind at the first submission) and execution (θ : solve rate given the first kind), median and 90% interval of a Bayesian bootstrap over instances. “sat.”: every session of every instance has (or lacks) the kind; the printed value is the smoothing floor rather than an estimate. A dash: fewer than three sessions in that cell.

E Adapter weights

The eleven LoRA adapters of this report are not deposited. Each weight file (`adapter_model.safetensors`, 318,843,352 bytes) has the SHA-256 in Table 13, recorded when the adapter was trained in `analysis/adapter_weights.sha256` and in the adapter manifests of the node’s evidence archive (`docs/a2c_evidence_v1` for A2, `docs/a2c2_evidence_v1` for the replication, `docs/a2c3_evidence_v1` and `docs/a2c4_evidence_v1` for the first two iterations of the loop, `docs/a9_evidence_v1` for A9), and checked against the local copies of the weights on 5 October 2026. The manifests also record the hashes of the training data, the adapter configuration and the training log, the seed, the number of steps and the library versions.

| adapter (node) | examples | seed | SHA-256 of <code>adapter_model.safetensors</code> |
|------------------------|----------|------------|---|
| p1 (A2, round 1) | 96 | 2026091301 | 27b83d449c85a89441efe844a1caa09781e8621839f64ab2ed35ab17270486fd |
| p2 (A2, round 1) | 288 | 2026091301 | d0182f3f9d6483e529ca33f3cfe2ba69bf805b34ddd7a98b8b27a680bb45abfe |
| p3 (A2, round 1) | 288 | 2026091301 | fd6876b389972e087a2775598421d919a1211c5894a5fca35ffc5f9dbb1b90ab |
| p1 (A2 replication) | 96 | 2026091402 | 6f1eef9e4869ce16bcde960d6359806d759f26898b5fc15531dc8e9fb1455f77 |
| p2 (A2 replication) | 288 | 2026091402 | 754a9bafeb417acdbd907ea73ee44bb5f3411785177c5a55c35e42c6673c3508 |
| p3 (A2 replication) | 288 | 2026091402 | 7db87a36029822cb3eefe90bb7dc64eb369908a9303fa5d770354d1f0fe0fad6 |
| own (iteration 1) | 60 | 2026091501 | 0d5679cc23dd48bb940f37406c02f37ed989d5fb8eaf4ee1fb7dfb254e76d664 |
| curm (iteration 1) | 60 | 2026091501 | 8a792395e3b30f4e3a4f313fb475879e4a2f5e9ceebebeca1fa966ed115c93370 |
| own2 (iteration 2) | 135 | 2026091502 | 3010271d6a9ae47dea171ed58d210136e487c6389535536d8e558ffbb3a37402 |
| curm2 (iteration 2) | 135 | 2026091502 | 5c2ca8b356c24e7a96d50ac19bdd59ca5932a548946d20a4a1128e83d02d65e7 |
| own3 (iteration 3, A9) | 187 | 2026091901 | 0d44c35369f0d60887c700d7297f74ecc01c0187334af21aee86e0bd936dff41 |

Table 13: The eleven adapters and the SHA-256 of their weight files, as recorded in the evidence archive of each node. Every adapter was trained for 108 steps with the recipe of Section 2.